

Sort Shapes Like a Mathematician

Answer Key

Grade 1

Quick Answers

- | | |
|--|---|
| 1. (a) how many 3-sided shapes = 2, — | 7. (a) 4-sided count = 5, (b) 6-sided count = 2, |
| 2. (a) how many 4-corner shapes = 3, — | (c) how many more = 3 |
| 3. (a) sides of each triangle = 3, — | 8. (a) corners of the odd one out = 3, — |
| 4. (a) 3-sided group = 2, (b) 4-sided group = 3, | 9. (a) equal sides of the tilted shape = 4, — |
| (c) shapes in neither group = 1 | 10. (a) 3-sided count = 3, (b) 4-sided count = 2, |
| 5. (a) 3-corner count = 4, (b) 4-corner count = | (c) 6-sided count = 2, (d) the self-check sum = |
| 2, (c) the comparison = > | 7 |
| 6. (a) shapes in all = 8, — | |
-

What is verified

4 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 1, 2, 3, 6, 8, 9. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 1

1.G – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Difficulty 1–4 of 5 across 25 tagged checks.

1. Shapes A (triangle), B (gray square), C (tilted triangle), D (rectangle). (a) How many have exactly 3 sides? (b) Letters?

Solution: (a) Count sides shape by shape, ignoring color and tilt: A has 3, B has 4, C has 3, D has 4. Two shapes have exactly 3 sides. 2

(b) *Instructor judges:* the letters are **A and C**.

2. Five shapes: B square, A triangle, D rectangle, E tilted square, F circle. (a) How many have exactly 4 corners? (b) Letters?

Solution: (a) B, D, and E each have 4 corners; A has 3; the circle F has none. That is three shapes. 3

(b) *Instructor judges:* the letters are **B, D, and E** — the tilted square still counts.

3. Big gray triangle, small white triangle. (a) Sides of each? (b) Is Kim right?

Solution: (a) Each shape has three straight sides.

3

(b) *Instructor judges:* Kim is not right. Both are triangles because both have 3 straight sides — size and color are not defining attributes, so the two shapes belong in the same group.

4. Six shapes sorted by sides. (a) 3-sided? (b) 4-sided? (c) Neither?

Solution: (a) The two triangles:

2

(b) The square, the tilted square, and the rectangle:

3

(c) Only the circle has no straight sides, so it fits neither group.

1

5. Six shapes sorted by corners. (a) 3 corners? (b) 4 corners? (c) $>$ or $<$?

Solution: (a) The four triangles:

4

(b) The square and the rectangle:

2

(c) Four is more than two, so

$4 > 2$

6. Raj's color sort: 3 gray shapes and 5 white shapes. (a) How many in all? (b) What should he count instead?

Solution: (a) Add the two groups: $3 + 5 = 8$.

8

(b) *Instructor judges:* color is not a defining attribute — Raj should sort by something countable that never changes, like the number of sides or corners.

7. The set with squares, rectangles, and pentagons. (a) 4-sided? (b) 5-sided? (c) How many more 4-sided?

Solution: (a) Two squares, two rectangles, and the tilted square:

5

(b) The two pentagons:

2

(c) Find the difference: $5 - 2 = 3$.

3

8. A square, B rectangle, C triangle, D tilted square. (a) Corners of the odd one out? (b) Its letter?

Solution: (a) A, B, and D all have 4 corners; the odd one out has 3.

3

(b) *Instructor judges:* the letter is **C** — the triangle.

9. Group rule: exactly 4 equal sides. Lily says the tilted shape is in the wrong group. (a) Equal sides? (b) Does it belong?

Solution: (a) Count them: the tilted shape has 4 equal sides.

4

(b) *Instructor judges:* yes, it belongs. It matches the rule exactly — 4 equal sides. “Looks like a diamond” is about how it sits, and position is not a defining attribute.

10. Challenge: seven shapes. (a) 3-sided? (b) 4-sided? (c) 6-sided? (d) Self-check sum?

Solution: (a) The three triangles:

3

(b) The square and the rectangle:

2

(c) The two hexagons:

2

(d) Add the group counts: $3 + 2 + 2 = 7$ — all seven shapes are sorted, none missed, none counted twice.

7
